



Gaming

Adobe Substance 3D Modeler — Interface, Concepts, Ecosystem and Use Cases

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What is Substance 3D Modeler

- A clay-first approach to 3D creation — sculpt naturally instead of building models point by point
- Traditional modeling: precise, programmatic, vertex/edge-based control
- Modeler's approach: intuitive, sculptural, similar to shaping physical clay
- Two sculpting modes: organic sculpting (smooth, freeform shapes) and hard surface sculpting (parametric primitives, boolean add/cut operations)
- Works seamlessly on both desktop and VR — switch between modes to match your project needs or preference
- Best suited for early-stage concept work, character sculpting, and rapid prototyping



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3D Modeler Interface





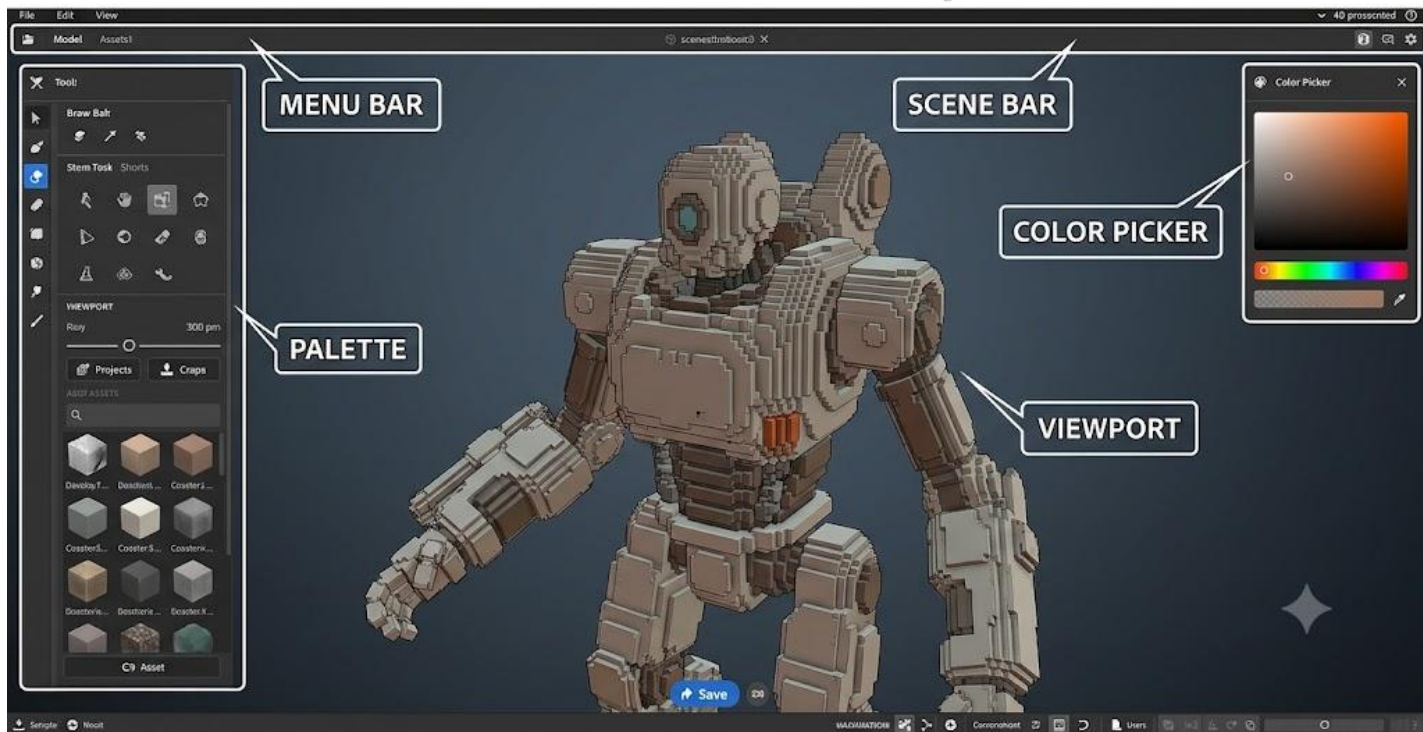
Interface Overview — Core Areas

- Menu bar — standard File, Edit, View, and Help menus
- Viewport — your working 3D space; desktop behaves like a camera (Alt+drag to orbit/pan/zoom), VR lets you physically move or use controller grips to navigate
- Palette — access all sculpting tools; press "F" on desktop to refocus camera on your sculpt if you get lost
- Color Picker — set clay color for the Clay and Paint tools; includes an Eyedropper to copy colors already in your scene
- Scene bar — quick scene-level controls and settings
- Action menu (hidden by default) — right-click on desktop for context-sensitive actions grouped by section



Interface Overview — Core Areas

Adobe Substance 3D Modeler: Desktop Interface Guide





The Palette — Structure

- Tool bar — select your active sculpting tool (Clay, Warp, Smooth, and more)
- Tool Parameters — fine-tune the selected tool's size, strength, and behavior; swap between tool variations here
- File Management — save and load your sculpt; note that some options stay locked until you save for the first time
- Placement mode — controls how your brush/tool interacts with the surface (e.g., along normal, freeform)
- Snapping controls — toggle angle or grid snapping for precision work, especially useful in hard surface sculpting



SUBSTANCE 3D MODELER: MODELER PALETTE PANEL

1. **TOOLS**
Brush, erase, and tools and tools

2. **ACTIONS**
Undo, enooiance, and boolean

3. **SCENE OBJECTS**
Geometry layers, ear6, geometry

4. **PROPERTIES**
Settings for selected tool/object

5. **ASSETS/COLOR** ✨
Materials, colors, and etc.

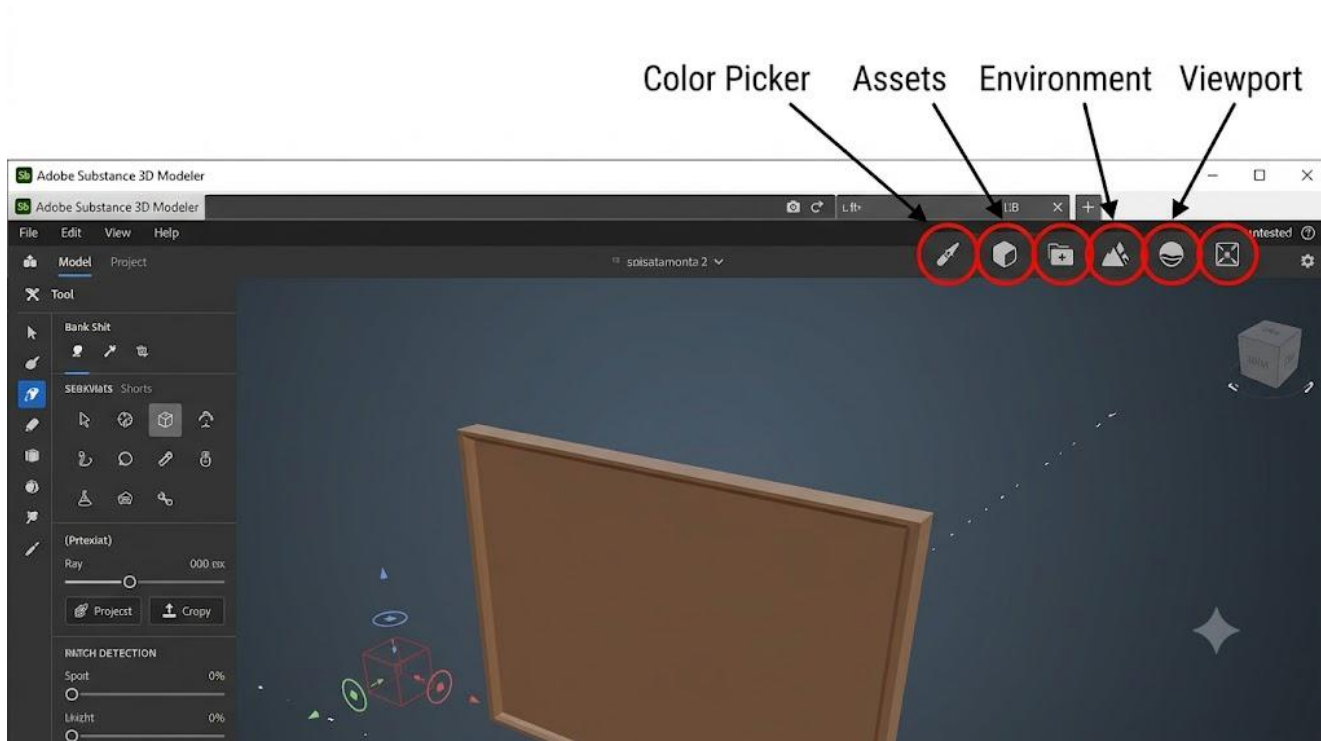


- Color Picker — always accessible in the top right; sets clay color for Clay and Paint tools only
- Materials panel — apply and manage surface materials on your sculpt directly from the top bar
- Assets panel — browse and drop in thousands of ready-made 3D assets from the Substance 3D library; imported assets come in as editable clay layers you can reshape with Modeler's tools
- Environment panel — control three-point lighting (key, fill, rim); adjust position, intensity, color, and toggle shadows per light
- Viewport panel — adjust how your scene is displayed, including background and viewport-level display settings



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Top Bar Panels





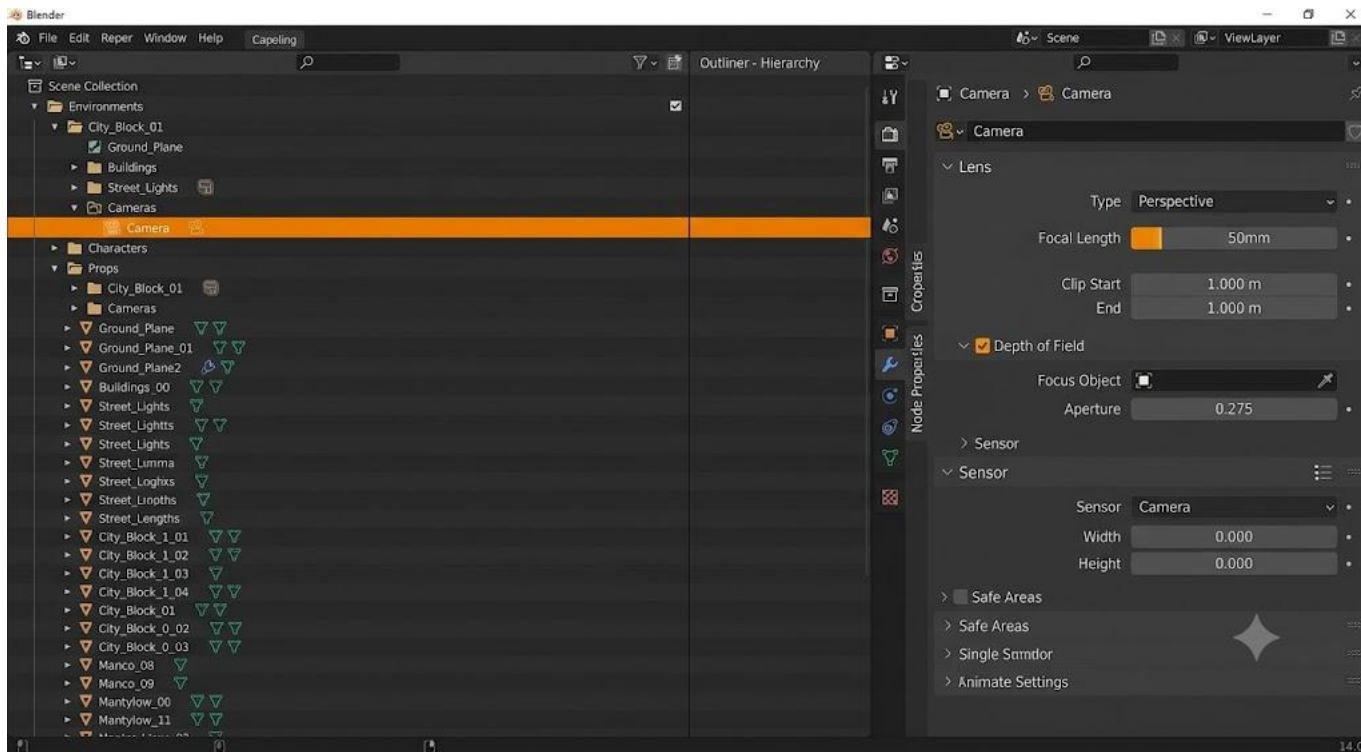
Outliner and Node Properties Panel

- Outliner — displays the full hierarchy of objects in your scene from a single location
- Drag objects into or out of groups directly within the Outliner; live visual cues show what happens before you drop
- Rename, rearrange, select, or delete objects from the Outliner without hunting through the viewport
- Useful for exporting multiple objects at once by changing your selection in the Outliner
- Node Properties panel — view and modify properties of a selected object; currently works only when a camera is selected
- Camera controls available: toggle perspective/orthographic view, adjust field of view, enable depth of field with blur amount, set a focus target, and change aspect ratio (portrait/landscape)
- Editing a selected camera does not affect your main viewport camera — the two are independent



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Outliner and Node Properties Panel





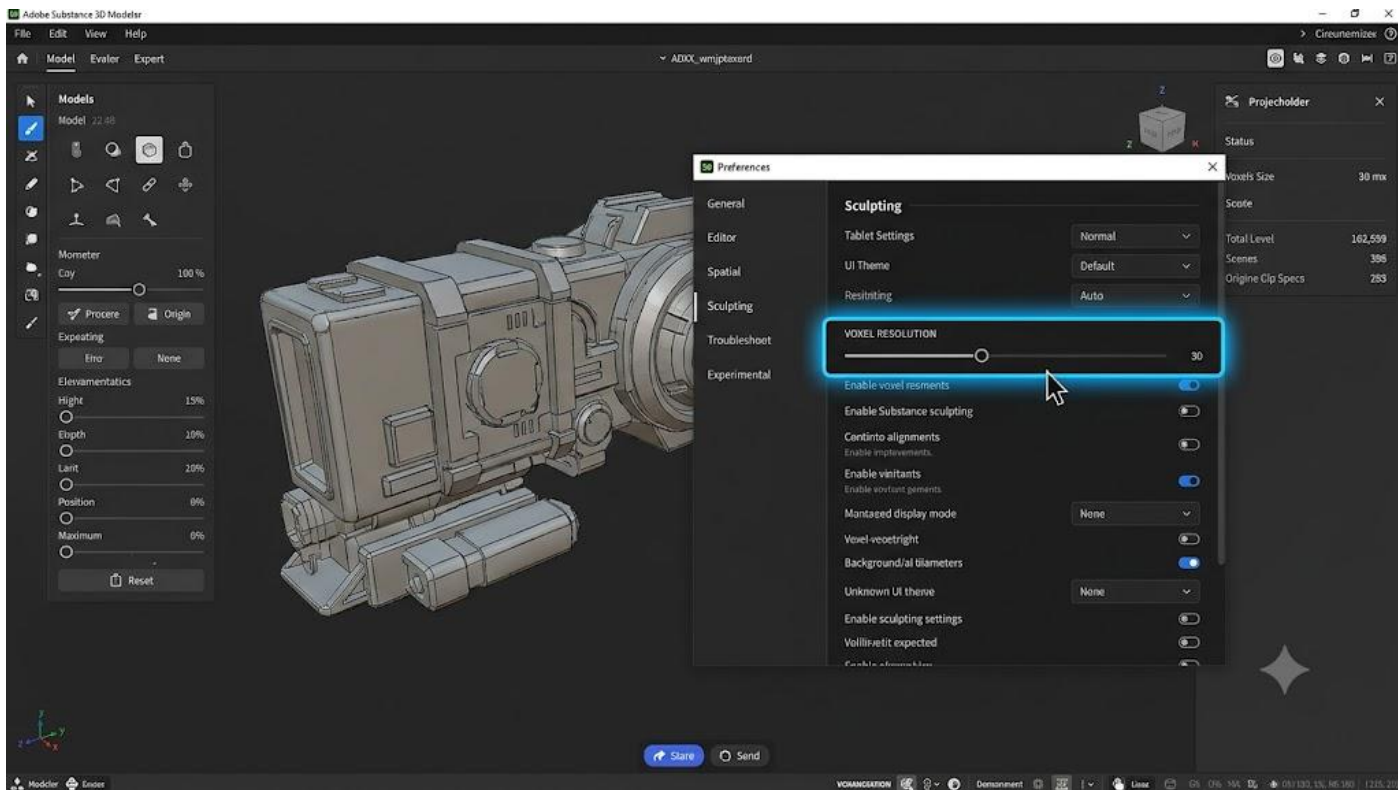
Bottom Bar and Preferences

- Bottom Bar left side — shows your current focus in the scene (what object/layer you're working on)
- Bottom Bar right side — houses commonly used actions for quick access without digging through menus
- Gizmo orientation controls — visible only in Gizmo placement mode; choose Local, Parent, or Global space
- Open Preferences via File menu or shortcut Ctrl+K
- Display preferences — toggle smooth shading, de-emphasize unfocused layers, show resolution on clay brush
- Performance preferences — enable max refresh rate override to reduce CPU/GPU usage on desktop; toggle ambient occlusion (off by default for performance)
- VR-specific preference — Letterbox display toggle, showing the true VR aspect ratio instead of the comfort-cropped desktop view



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Bottom Bar and Preferences





Desktop vs VR Mode

- Same features everywhere — nothing is locked to one platform; desktop and VR are treated as equals
- Desktop input — mouse/pen with keyboard; cursor moves like a normal 2D pointer
- VR input — any supported 6DOF controller; cursor sits at the tip of your tool-hand controller
- Tool hand vs support hand — dominant hand controls tools, non-dominant hand supports/assists (grips, menus)
- Entering/exiting VR — simply put on or remove the headset; desktop window mirrors your VR view live
- Best for VR — organic, freeform shapes (curves, wobbles, character sculpting) using natural hand movement
- Best for desktop — precise, mechanical, hard-surface objects (architecture) using traditional gizmos
- Headless mode (experimental) — use VR controllers on desktop without wearing a headset. useful for motion-sickness-prone users



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Desktop vs VR Mode





Building Blocks of 3D — Vertices, Edges, Faces

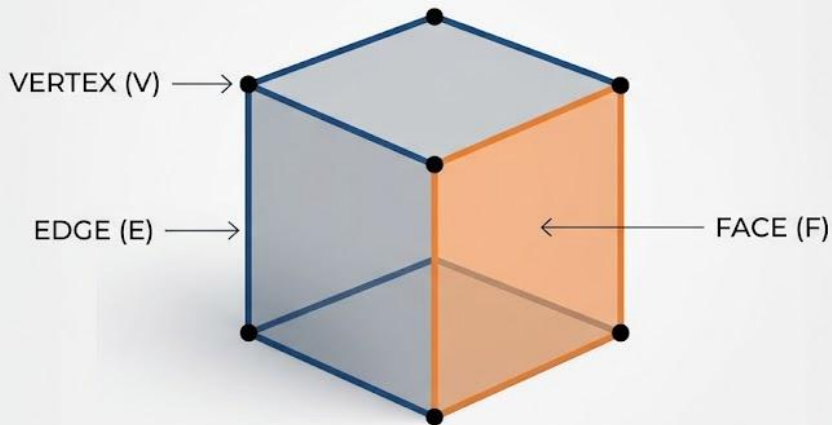
- Vertex — a single point in 3D space, defined by X, Y, Z coordinates; the simplest possible element
- Edge — a straight line connecting exactly two vertices; forms the "wireframe" skeleton of a model
- Face (polygon) — a flat surface bounded by a closed loop of edges (minimum 3); this is what actually gets shaded and rendered
- Order of construction: vertices → edges → faces — each level builds on the one before it
- A cube example: 8 vertices, 12 edges, 6 faces
- Wireframe view shows only vertices and edges (no shading); solid/rendered view shows the filled faces
- Edges and vertices are invisible in the final render — only faces produce the visible surface



Building Blocks of 3D — Vertices, Edges, Faces

STRUCTURE OF A CUBE

Labeled Diagram



V: Vertex | **E:** Edge | **F:** Face



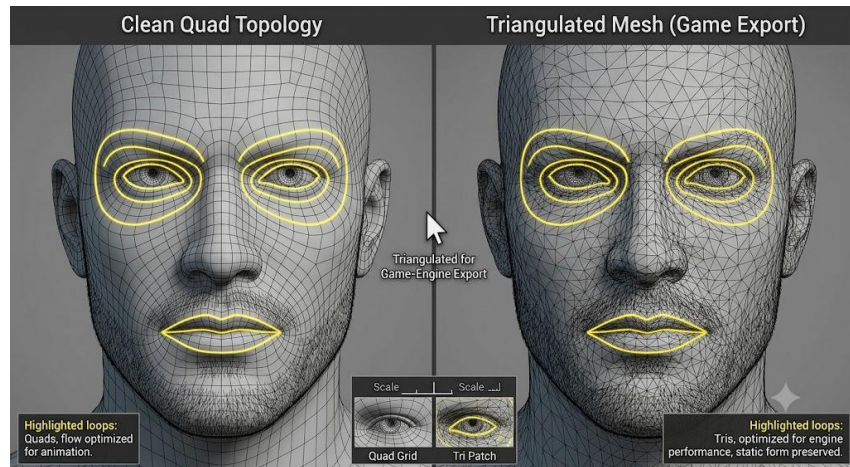
Polygons and Meshes

- Mesh — the complete collection of vertices, edges, and faces that together define an object's 3D shape
- Triangle (tri) — 3 vertices, 3 edges; simplest possible polygon; GPUs render everything as triangles internally
- Quad — 4 vertices, 4 edges; preferred by artists for modeling because it creates clean edge loops and subdivides smoothly
- Edge loops — continuous rings of quad edges that follow natural contours (like around a mouth or joint); critical for smooth animation and deformation



Polygons and Meshes

- Quads for artists, triangles for engines — model and edit in quads, but real-time engines like Unity and Unreal automatically triangulate on import/export
- Wireframe view — displays only the mesh structure (vertices, edges) without solid shading, useful for checking topology
- Poly count — total number of polygons in a mesh; low poly (hundreds to a few thousand) suits real-time/mobile use, high poly (tens of thousands+) suits cinematic detail





Primitives in 3D Modeling

- Primitive — a simple, pre-defined 3D shape used as a starting point for modeling
- Standard primitives — the essentials: Box/Cube, Sphere, Cylinder, Cone, Torus, Pyramid, Plane
- Extended primitives — advanced variants with more parameters, e.g. ChamferBox (rounded-edge box), Capsule, Oil Tank (cylinder with rounded caps), Hedra
- Each primitive comes with adjustable parameters — radius, height, segments, smoothness — that shape its final form before further editing
- Choosing the right primitive saves work — pick the shape closest to your end result to minimize editing steps (e.g. cylinder for a mug, sphere for a ball)
- Segments matter — increasing segment count on a sphere/cylinder gives smoother curves but adds more polygons to manage
- Box modeling — starting from a cube and using extrude, loop cut, and scale to gradually sculpt toward a complex final shape



Primitives in 3D Modeling

3D Primitive Shapes and Extended Counterparts Comparison



Box



Chamfer Box



Sphere



Capsule



Cylinder



Oil Tank



Cone



Extended
Cone



Torus



Extended
Torus



Pyramid



Extended
Pyramid



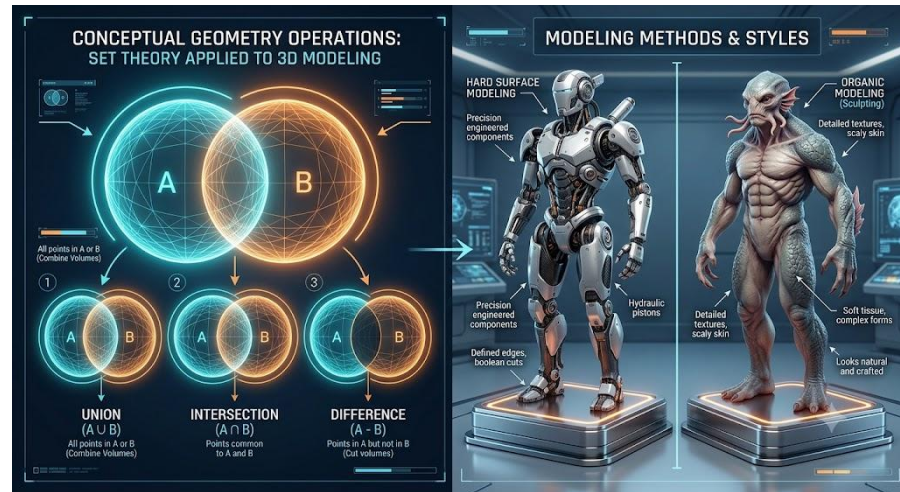
Organic Sculpting vs Hard Surface Sculpting

- Organic sculpting — used for characters, creatures, plants, rocks, food; anything with soft, irregular, natural forms
- Hard surface sculpting — used for machines, vehicles, buildings, weapons; objects with sharp edges, straight lines, and mathematically precise curves
- Organic workflow — additive/subtractive clay-based sculpting with brushes, focused on form, flow, and artistic feel rather than exact measurement
- Hard surface workflow — built from primitives combined using boolean operations, focused on precision and predictable geometry
- Boolean Union — merges two objects into one combined shape (addition)



Organic Sculpting vs Hard Surface Sculpting

- Boolean Difference/Subtract — cuts one object away from another, creating holes or notches
- Boolean Intersection — keeps only the overlapping volume between two objects, discarding the rest
- Boolean Split — divides a target object into separate pieces based on where a second object's surface cuts through it
- In Modeler specifically — select a layer first before applying Union to add clay to it; with nothing selected, Union creates a new layer instead

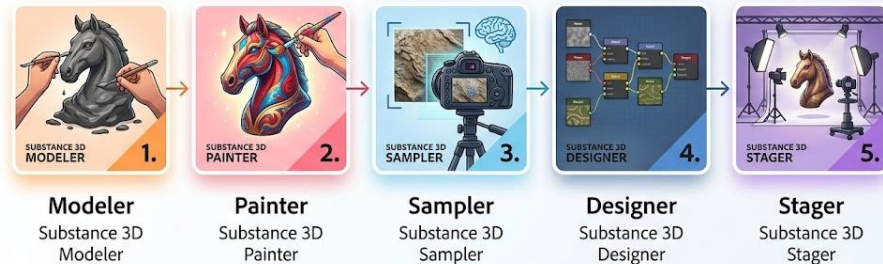




The Substance 3D Suite — Introduction

- Substance 3D Modeler — sculpts and creates 3D models using a clay-like, intuitive workflow on desktop and VR (Windows only)
- Substance 3D Painter — paints realistic textures onto 3D models in real time using advanced brushes and Smart Materials
- Substance 3D Sampler — turns real-life photos into custom, tweakable materials for use across 3D projects
- Substance 3D Designer — the industry-standard, node-based procedural tool for building materials and filters with total artistic control

SUBSTANCE 3D WORKFLOW PIPELINE





The Substance 3D Suite — Introduction

- Substance 3D Stager — composes and renders finished 3D scenes, acting like a virtual photography studio with lighting and camera setup
- Substance 3D Assets — a library of nearly 20,000 ready-to-use materials, models, and HDR lights available across all apps
- Typical pipeline order: Model (Modeler) → Texture (Painter/Sampler) → Compose & Render (Stager)
- All apps are deeply interoperable — assets and materials move freely between them within Creative Cloud



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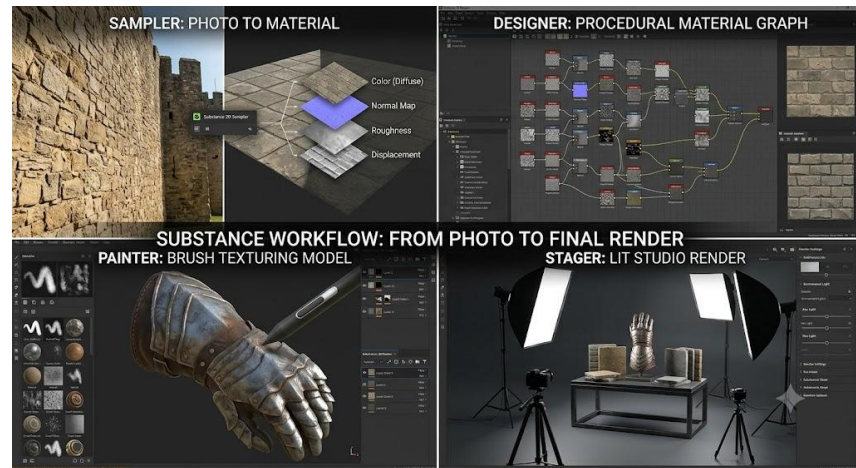
Substance 3D Designer, Sampler, Painter & Stager

- Substance 3D Designer — node-based, procedural material creation; builds materials from scratch using filters, patterns, and noise, giving infinite tweakable variations
- Designer key features — huge content library of nodes/filters, HDR lighting creation, Pantone/OpenColorIO color management, MDL (material definition language) support for realistic shaders
- Substance 3D Sampler — turns real-world photos or scans into custom, editable materials using parametric filters applied in a layer stack
- Sampler-to-Designer link — export a Sampler material to Designer to get a head start on its node graph instead of building from zero



Substance 3D Designer, Sampler, Painter & Stager

- Substance 3D Painter — paints textures directly onto 3D models in real time using brushes and Smart Materials that adapt to a model's geometry
- Substance 3D Stager — composes finished 3D scenes like a virtual photography studio; handles layout, lighting, camera angles, and final rendering
- Simple role split — Designer/Sampler create materials, Painter applies them to a model's surface, Stager assembles and renders the finished scene





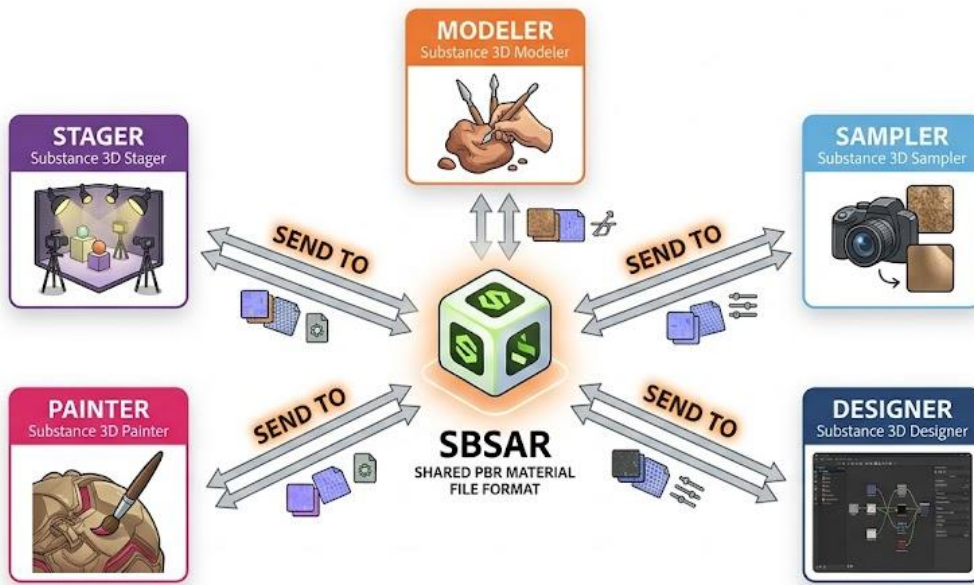
Ecosystem Integration — Send-To Workflow

- SBS file — the fully editable source file, only openable in Substance 3D Designer with full node-graph access
- SBSAR file — a packaged, portable archive exposing only chosen parameters; works across all Substance apps and most game engines/DCC
- Send To feature — built into Sampler, Designer, Painter, and Stager; one click packages an asset with correct settings for the target
- Sampler → Designer/Painter/Stager — send a finished material directly for further refinement or texturing
- Painter → Stager/Photoshop/After Effects — send a fully textured model for final scene composition or 2D compositing



Ecosystem Integration — Send-To Workflow

SUBSTANCE 3D ECOSYSTEM: SBSAR HUB-AND-SPOKE WORKFLOW



PBR MATERIALS, TEXTURES, AND GEOMETRY SHARING



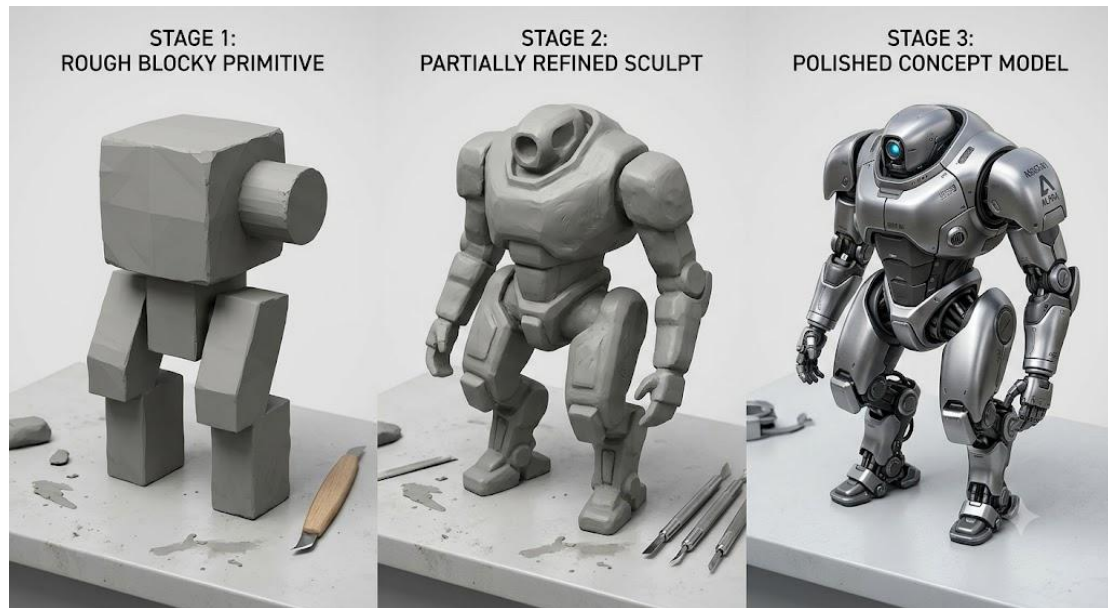
Concept Sculpting Workflow

- Purpose — quickly explore design ideas and visualize forms in 3D before committing to a final model
- SDF (Signed Distance Field) engine — Modeler's core technology lets you sculpt freely without worrying about polycount, topology, or subdivision levels while concepting
- Start broad, refine later — block out the overall silhouette with large brushes and primitives first, add fine detail only once the shape feels right
- "Eyeballing it" is fine — concepting favors visual judgment over exact measurements; accuracy comes in later production stages
- Use symmetry — mirror or radial symmetry speeds up exploring balanced, complex forms (e.g. a product with matching side panels)
- Iterate non-destructively — keep booleans and layers editable so you can adjust or undo design decisions without starting over



Concept Sculpting Workflow

- VR advantage for concepting — sculpt at life-size scale with natural hand movement, useful for visualizing how a design feels in real space
- Real-world use — product design concepts (e.g. a toaster shape), character/creature blockouts, architectural massing studies





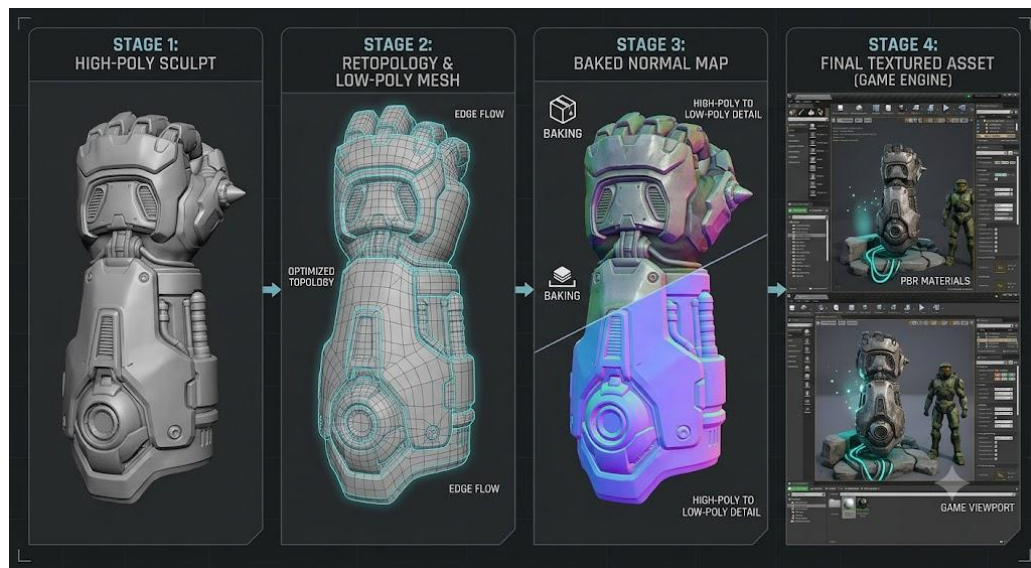
Asset Creation for Games and Film

- Symmetry and repetition tools — build complex, detailed objects efficiently by mirroring or repeating shapes instead of sculpting each side manually
- Hard surface + organic combo — most game/film assets mix both: hard surface for mechanical parts, organic sculpting for surface wear, damage, or creature-like details
- Export mode (Ctrl+Shift+E) — Modeler lets you export the whole scene, your current selection, or a single defined Asset depending on project needs
- Typical downstream pipeline — concept/blockout → high-poly sculpt (Modeler) → retopology (clean low-poly mesh) → UV unwrapping → baking → texturing (Painter) → engine import



Asset Creation for Games and Film

- Retopology — rebuilding a high-detail sculpt into an efficient low-poly mesh suitable for real-time rendering in Unity/Unreal
- Baking — transferring fine sculpted detail from the high-poly model onto normal maps applied to the low-poly version, preserving detail without the heavy polycount
- Common export format — FBX is the industry standard for moving assets between 3D software and game engines





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**System requirements and
installation, VR vs. Desktop
workflows (hardware overview)**

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Why System Requirements Matter

- VR applications must render immersive 3D scenes in real time.
- A low-performance system can cause lag, frame drops, poor visuals, or application crashes.
- In VR, poor performance may lead to motion sickness, discomfort, and reduced user immersion.
- System requirements affect development speed, build time, testing quality, and final application performance.
- Developers must consider the requirements of both the development computer and the target VR device.

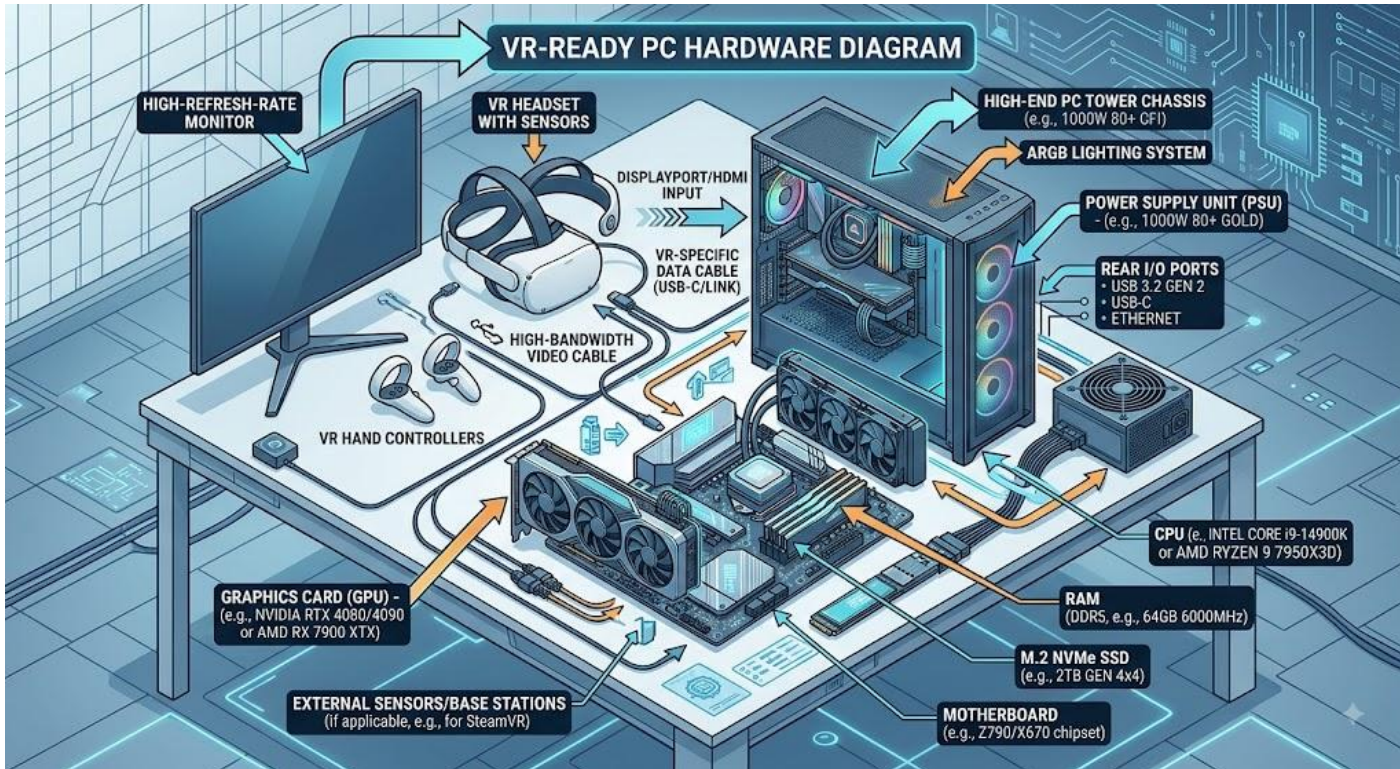


Hardware Components for VR Development

- CPU: Executes game logic, physics, AI, input processing, and background tasks.
- GPU: Renders 3D models, textures, lighting, shadows, and stereoscopic VR views.
- RAM: Temporarily stores assets, scenes, textures, and running application data.
- Storage: SSD storage improves project loading, asset import, and build time.
- VR headset: Displays the virtual environment and tracks head movement.
- Motion controllers: Allow hand-based interaction with virtual objects.
- Audio device: Headphones or spatial-audio hardware provide immersive sound.
- Tracking sensors/cameras: Track position and orientation in room-scale VR systems.



Hardware Components for VR Development



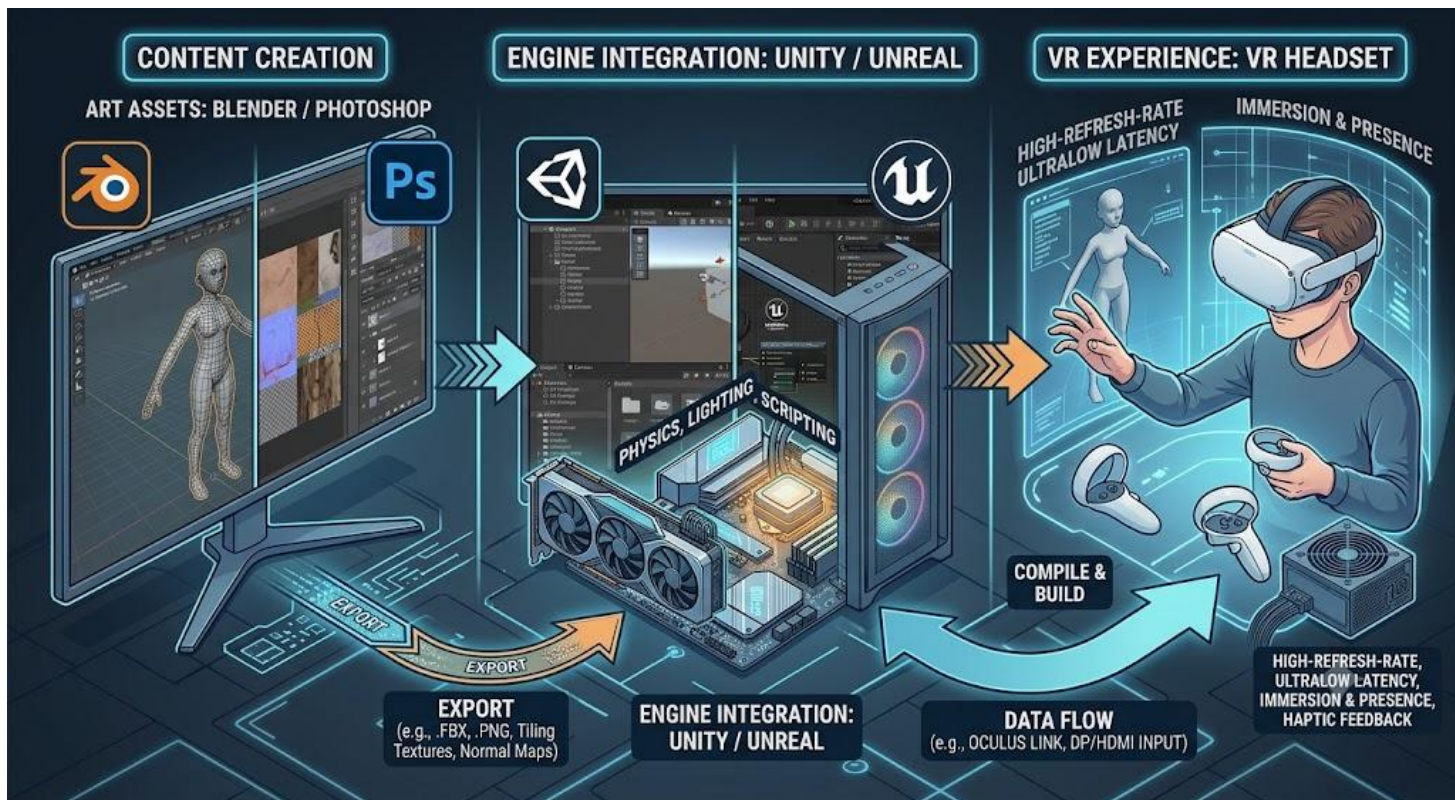


Software Requirements and Development Tools

- Operating system: A compatible version of Windows, macOS, Linux, or the headset operating system.
- Game engine: Unity or Unreal Engine for designing scenes, interaction, physics, and deployment.
- VR platform SDK: Platform-specific tools for headset integration, tracking, input, and deployment.
- IDE or code editor: Visual Studio, Visual Studio Code, Rider, or another compatible editor.
- Graphics drivers: Updated GPU drivers improve stability and compatibility.
- 3D asset tools: Blender, Maya, 3ds Max, Substance 3D, or similar tools for models and textures.
- Version control: Git and GitHub/GitLab for managing project files and teamwork.



Software Requirements and Development Tools





Gaming Development Setup Tiers Using Adobe Substance 3D Tools

- **Basic setup:** Modern CPU, 8 GB RAM, dedicated GPU, SSD, Full HD monitor, Blender, Substance 3D Painter, and Unity/Unreal Engine.
- **Recommended setup:** Multi-core CPU, 16 GB RAM, powerful GPU, NVMe SSD, and Substance 3D Painter, Sampler, Stager, and game engine tools.
- **Advanced VR/game setup:** High-performance CPU, 32 GB RAM, high-end GPU with good VRAM, large SSD storage, VR headset, controllers, and Unity/Unreal XR tools.
- **Substance 3D role:** Create realistic textures, materials, and environment assets for games.
- **Game-engine role:** Import Substance assets into Unity or Unreal Engine for gameplay, lighting, interaction, and deployment.
- **Key requirement:** Optimize models, textures, and lighting to maintain smooth performance, especially in VR.



Pre-Installation Checklist

- Check CPU, GPU, RAM, SSD space, and operating-system compatibility.
- Update graphics drivers and operating system.
- Install required game-development and Adobe Substance 3D tools.
- Connect and update VR headset and controllers, if used.
- Create folders for Assets, Textures, Models, Scenes, Scripts, and Builds.
- Keep backup storage for game projects and exported assets.

POWERFUL WORKSTATION: High-end CPU (e.g., i9/Ryzen 9), ample RAM (e.g., 64GB+), and efficient cooling.

PERFORMANCE GPU: Modern GPU (e.g., RTX 4080/4090) for real-time rendering, Physics, and 90+ FPS.

VR HEADSET SYSTEM: PC-VR compatible headset, motion controllers, and sensor base stations.

CONTENT CREATION TOOLS: 3D Modeling (e.g., Blender) & 2D Texture Creation (e.g., Photoshop).

ENGINE INTEGRATION: Game/experience engine (e.g., Unity or Unreal) for assembly and logic.



Desktop Gaming Workflow

- Game is displayed on a monitor.
- Player uses keyboard, mouse, or game controller.
- Camera is controlled through mouse movement or controller input.
- UI uses menus, buttons, HUD, health bars, and score panels.
- Testing is done using the game-engine preview window or desktop build.
- Suitable for PC games, 2D games, and non-immersive 3D games.





VR Gaming Workflow

- Game is experienced through a VR headset.
- Player looks around by moving their head.
- Motion controllers or hand tracking enable natural interaction.
- Objects can be grabbed, moved, thrown, or activated.
- UI is placed inside the 3D virtual environment.
- Testing must be done inside the headset for comfort and usability.





VR vs. Desktop Workflow

Feature	Desktop Workflow	VR Workflow
Display	Monitor	VR headset
Input	Keyboard, mouse, controller	Motion controller, hand tracking, gaze
Camera	Controlled by player input	Controlled by head movement
View	Single screen view	3D stereoscopic view
UI	Screen-based HUD and menus	World-space virtual UI
Testing	Desktop preview/build	In-headset testing
Main focus	Gameplay and graphics	Immersion, comfort, and interaction



Interaction and User Experience

- Desktop games use clicks, keys, buttons, and menus.
- VR games use physical movement, gestures, pointing, and grabbing.
- VR interaction should feel natural and easy to understand.
- Important VR features include spatial audio, realistic scale, and clear feedback.
- Keep important objects within a comfortable reach of the player.
- Use readable text and avoid placing UI too close to the user.



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Answer the following

1. What are the basic system requirements for game and VR development?
2. State the role of Adobe Substance 3D Painter in game development.
3. Differentiate between desktop and VR workflows.
4. Why is optimization important in game development?
5. List any four common VR hardware components.



Gaming

Viewports: Object space and world space, Brush menu, tool shelf, and layer panel

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Substance 3D Modeler Interface and Workspace

Adobe Substance 3D Modeler is a 3D sculpting application used to create models with digital clay and primitives. It supports desktop-based modeling and virtual reality (VR) modelling. The interface helps artists create, modify, organize, and inspect 3D objects efficiently.





Viewport: The Main 3D Workspace

- The viewport is the large central 3D work area in Substance 3D Modeler.
- It is the space where users create and edit clay objects, primitives, layers, and complete scenes.
- Anything that is not part of the interface controls is generally part of the viewport.
- On a desktop, the viewport behaves like a camera looking into a 3D environment.
- In VR, the user experiences the viewport as an immersive space surrounding them.



Navigating the Viewport in Desktop and VR

Desktop viewport navigation:

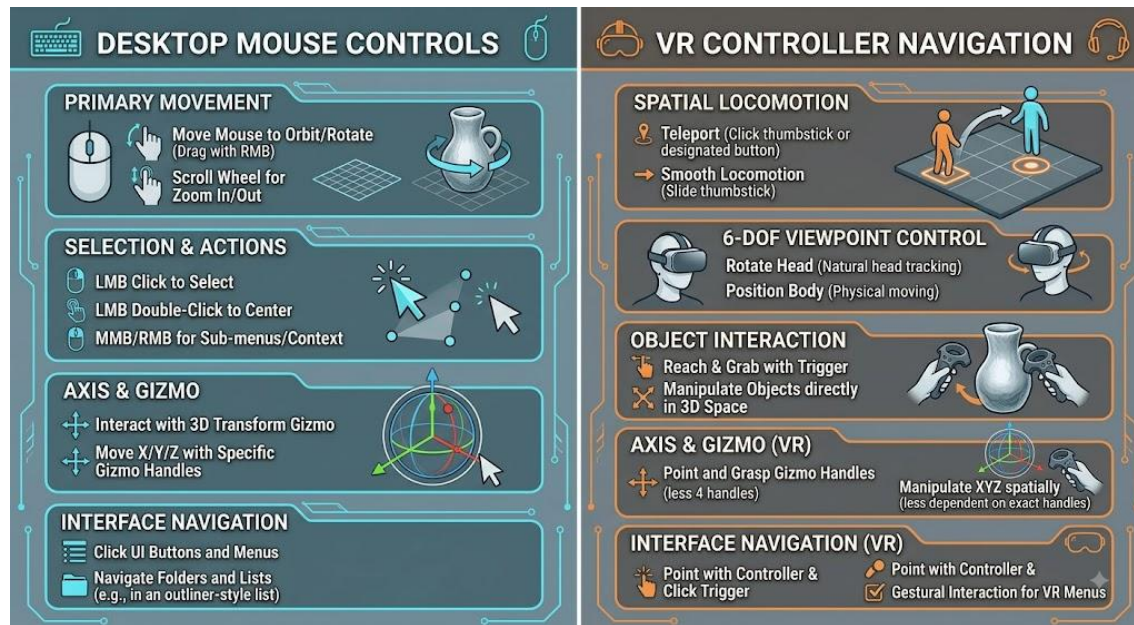
- Orbit/Rotate: Changes the camera direction around the model.
- Shortcut: Alt + left mouse drag.
- Pan: Moves the camera horizontally or vertically.
- Shortcut: Alt + middle mouse drag.
- Zoom: Moves the camera closer to or farther from the model.
- Shortcut: Alt + right mouse drag.
- Focus: Press F to bring the camera focus back to the sculpt or gizmo.



Navigating the Viewport in Desktop and VR

VR viewport navigation:

- Walk physically around the scene.
- Hold one controller grip and move the controller to reposition the scene.
- Hold both grips and move controllers closer or farther apart to zoom.





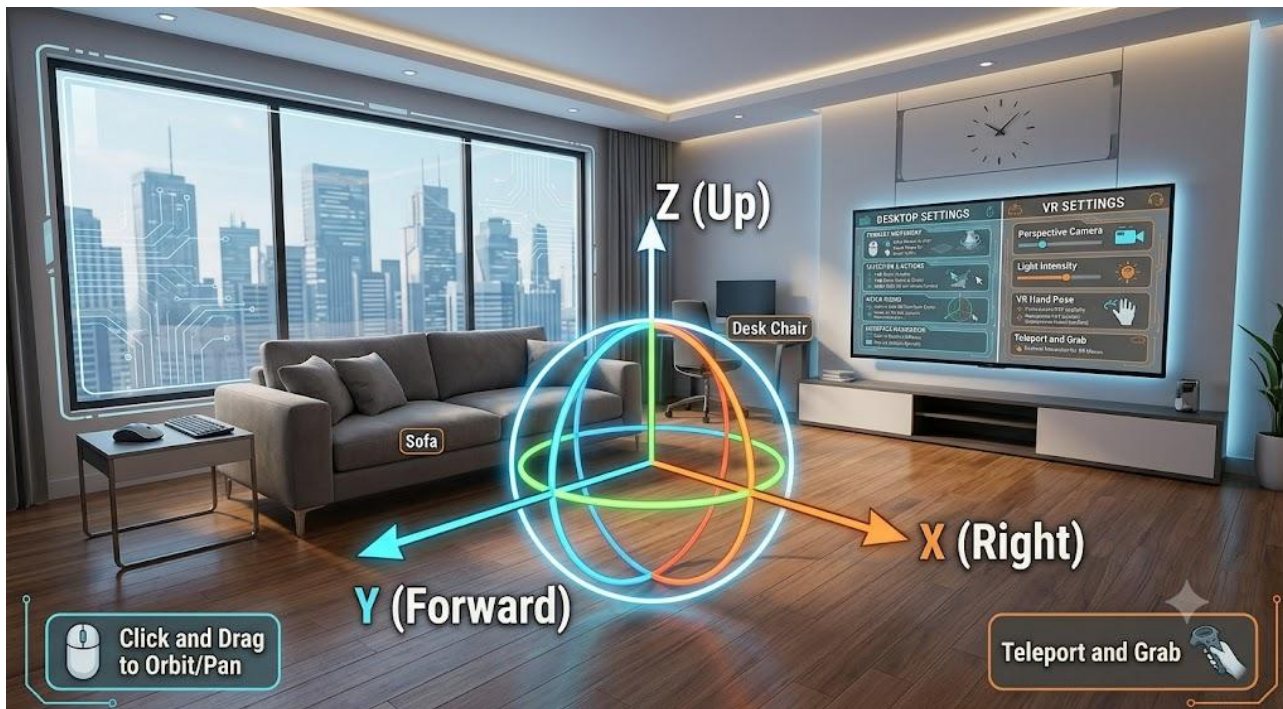
Understanding World Space

- World space is the global coordinate system of the complete 3D scene.
- It provides a fixed reference frame for every object in the scene.
- The origin of world space is generally represented as 0,0,0.
- The axes are commonly represented as:
 - X-axis: Left and right direction
 - Y-axis: Up and down direction
 - Z-axis: Forward and backward direction
- World space is useful when placing multiple objects in a common scene.



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Understanding World Space





Understanding Object Space

- Object space is the local coordinate system of an individual object.
- Its axes move and rotate together with that object.
- Object space is useful when editing details based on the object's own direction.
- For example, after rotating a chair, its local “front” direction rotates with the chair.
- Object space is especially important for transformations, local editing, animation, and precise placement of details.



Understanding Object Space





Object Space vs World Space

Feature	World Space	Object Space
Reference point	Entire scene	Individual object
Axis direction	Fixed for the scene	Rotates and moves with the object
Best used for	Arranging objects in a scene	Editing or transforming a selected object
Example	Move a chair toward the room's north wall	Move a rotated chair forward according to its own front
Coordinate origin	Scene origin	Object's local origin



Brush Menu and Brush Parameters

- A brush determines where and how a selected tool affects clay.
- With the Clay tool, the brush decides the shape and location of added clay.
- Almost every sculpting tool uses a brush; the Select tool is an important exception.
- Brush behavior can be adjusted from the parameters section of the Palette.





Brush Menu and Brush Parameters

Common brush parameters:

- Brush shape: Selects the basic form, such as sphere, cube, or torus.
- Scale: Increases or decreases brush size.
- Pressure: Allows pressure-sensitive variation in brush size.
- Single mode: Places one stamp instead of a continuous stroke.
- Lazy stroke: Smooths strokes by making them trail behind the cursor.
- Steady stroke: Smooths movement by making the stroke gradually catch up to the cursor.

Practical tip:

- Use a large brush for rough forms and a small brush for facial features, edges, grooves, and fine details.



Tool Shelf: The Modeler Palette

- In Substance 3D Modeler, the tool shelf is officially called the Palette.
- The Palette is one of the most frequently used parts of the interface after the viewport.
- On desktop, it is generally visible in the top-left area of the application window.
- It provides access to tools, tool galleries, and settings for the currently active tool.
- Users can change its placement through Preferences > Desktop > Handedness.



Tool Shelf: The Modeler Palette

Main Palette sections:

- Tools section: Select a sculpting or editing tool.
- Tool gallery: Access related tools within a category.
- Tool parameters: Modify the behavior of the selected tool and brush.
- File controls: Available at the bottom of the tools area.



Tool Galleries and Common Sculpting Tools

- A tool gallery is a group of related tools.
- Selecting a tool from a gallery reveals related tools in the Palette.
- This keeps the interface organized and reduces clutter.

Examples of tool galleries:

Warp gallery:

- Warp tool: Pushes or deforms clay.
- Elastic tool: Stretches clay with an elastic effect.

Erase gallery:

- Erase tool: Removes clay.
- Split tool: Separates clay into portions.
- Crop tool: Cuts away selected regions.



Tool Galleries and Common Sculpting Tools

Buildup gallery:

- Buildup tool: Adds and builds clay volume.
- Crease tool: Creates grooves, folds, or sharp surface lines.

Other useful tools:

- Clay tool for adding material.
- Inflate tool for expanding surfaces.
- Paint tool for applying color.



Layers, Scope, and Scene Organization

- Layers are fundamental scene objects in Substance 3D Modeler.
- A layer is a container that holds clay; without layers, a Modeler scene cannot hold clay.
- Use separate layers to organize different parts of a model.

Example for a character model:

- Layer 1: Head
- Layer 2: Neck
- Layer 3: Torso
- Layer 4: Arms or accessories



Layers, Scope, and Scene Organization

Layer operations:

- Create a layer from the Actions menu.
- Quickly create a layer with Ctrl + N on desktop.
- Duplicate a layer with Ctrl + D.
- Change layer resolution when more or less surface detail is needed.
- Use linked instances when repeated objects should update together.

Scope:

- Scope controls the level of the hierarchy currently being edited.
- When scoped into a layer, edits affect only that layer.
- On desktop, double-click empty space to scope out.
- Use the Select tool and double-click an object to scope in.



Workflow Example

- Workflow example: Creating a simple clay robot
- Open a new scene and create separate layers for body, head, arms, and legs.
- Use the viewport controls to orbit, pan, zoom, and focus on the model.
- Use large Clay or Buildup brushes to block the main body shape.
- Use smaller brushes and the Crease tool to add eyes, joints, and surface details.
- Use object-oriented thinking to shape each part according to its local direction.
- Use world-space thinking to place all robot parts correctly in the complete scene.
- Organize, rename, duplicate, or instance layers where symmetry or repeated parts are needed.



Gaming

**Understanding sculpting modes and
transform controls
Navigation shortcuts and layout
organization**

Shambavi BN

Department of Computer Applications



Introduction to Sculpting in Adobe Substance 3D

Sculpting in Adobe Substance 3D involves digitally molding, carving, and refining a 3D model, similar to shaping clay by hand. It enables artists to build organic forms, fine surface details, and realistic textures that make models more visually compelling.

- Supports extensive high-detail customization
- Gives intuitive, hands-on control over a model's surface
- Plays an important role in creating realistic characters, props, and environments





Overview of Sculpting Modes in Adobe Substance 3D

Adobe Substance 3D provides multiple sculpting modes that determine how a brush modifies a model's surface.

- **Additive:** Builds up material to form raised details and volumes.
- **Subtractive:** Removes material to carve grooves, dents, or recessed areas.
- **Smooth:** Blends and softens uneven regions for a cleaner, more polished surface.
- **Flatten:** Evens out areas to create flat surfaces or more defined edges.
- **Other modes:** Tools such as Pinch and Inflate produce specialized effects, including tightening surface areas or expanding forms.

Artists can switch between these modes throughout the sculpting workflow to achieve the





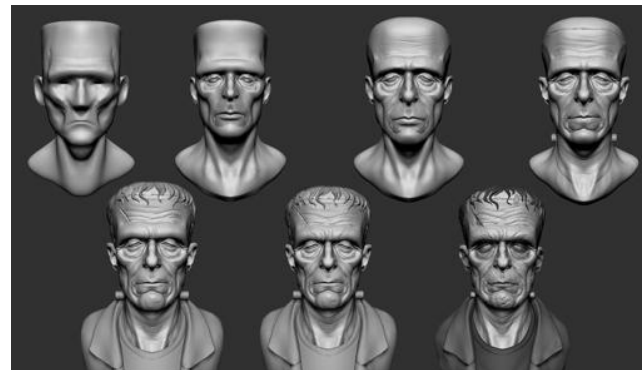
Sculpting Mode: Additive and Subtractive

Additive Mode

Adds material to create raised details such as muscles, wrinkles, bark, and cloth folds. Brush size and strength control the intensity of the effect.

Subtractive Mode

Removes material to form dents, creases, cracks, scars, and engraved details. Depth and softness can be adjusted for a natural finish.





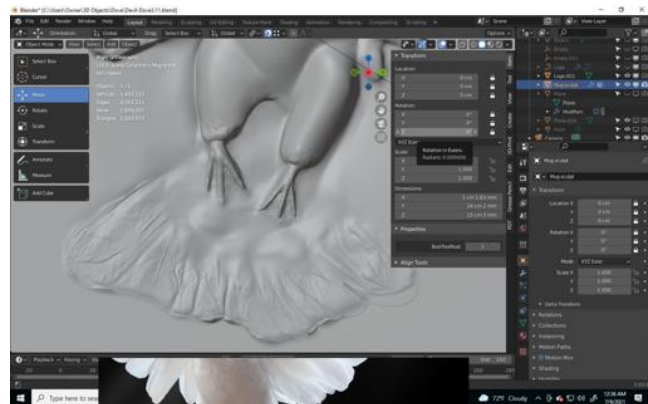
Sculpting Mode: Smooth and Flatten

Smooth Mode

Softens rough areas and blends sculpted details for natural transitions. Adjust brush strength to control the amount of smoothing.

Flatten Mode

Levels uneven surfaces to create flat planes and sharper shape definition. It is useful for mechanical objects, hard-surface models, and stylized characters.





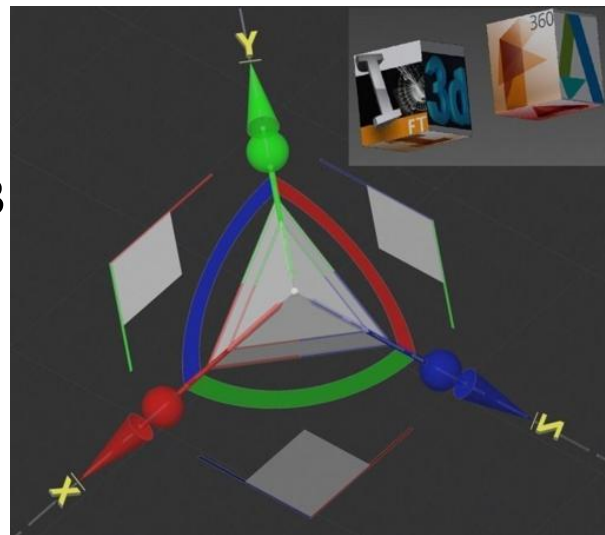
Introduction to Transform Controls in Adobe Substance 3D

Transform Controls

Transform tools modify the overall position, orientation, and size of a 3D model or selected area.

- **Move:** Repositions the model or selected part in 3 space.
- **Rotate:** Changes its orientation around an axis.
- **Scale:** Enlarges or reduces its size uniformly or along specific axes.

These controls help adjust proportions and poses before adding fine sculpted details, without unintentionally changing the surface work.





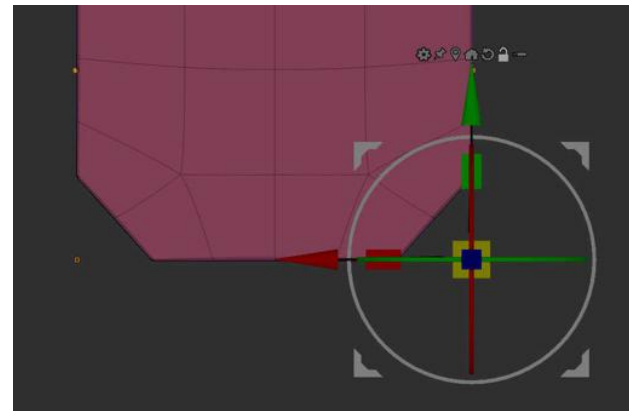
Using Transform Controls in Sculpting

Applying Transform Controls

Select the complete model or a specific region, then use the transform gizmo to move, rotate, or scale it. This helps reposition sculpted areas and refine proportions before finalizing the form.

Precision Tips

Use axis locks and snapping for accurate movement. Apply masking to transform only selected regions. Adjust the pivot point for more natural rotations or precise mechanical movement.





Navigation Overview in Substance 3D

Efficient navigation helps you inspect and edit models from any angle in Substance 3D.

- **Orbit:** Rotates the camera around the object.
- **Pan:** Moves the view horizontally or vertically.
- **Dolly (Zoom):** Moves the camera closer to or farther from the object.

Navigation can be performed using mouse buttons and keyboard shortcuts for faster, more precise viewport control.





Essential Navigation Shortcuts (Mouse)

Use these shortcuts to move around the 3D viewport in Substance 3D Modeler:

Action	Shortcut
Orbit	Hold Alt (Option on Mac) + drag with the Left Mouse Button
Pan	Hold Alt (Option on Mac) + drag with the Middle Mouse Button
Dolly / Zoom	Hold Alt (Option on Mac) + drag with the Right Mouse Button

These controls let you rotate around the model, shift the camera view, and zoom in or out smoothly. The camera movement is relative to the gizmo or the cursor position.



Essential Keyboard Navigation Shortcuts

Keyboard shortcuts make viewport navigation faster and more precise.

- **Frame Selected:** Press F to center the selected object in the viewport.
- **Reset View:** Press Shift + F to return the camera to its default view.
- **Switch Workspaces:** Use Ctrl + 1, Ctrl + 2, Ctrl + 3, and so on (Cmd on Mac) to switch workspace layouts.
- **Orbit with Numpad:** Use 4, 6, 8, and 2 on the numeric keypad to orbit left, right, up, and down in fixed steps.





Workspace Layout and Panel Organization

Substance 3D Workspace

The Substance 3D workspace contains key areas that support an efficient 3D workflow:

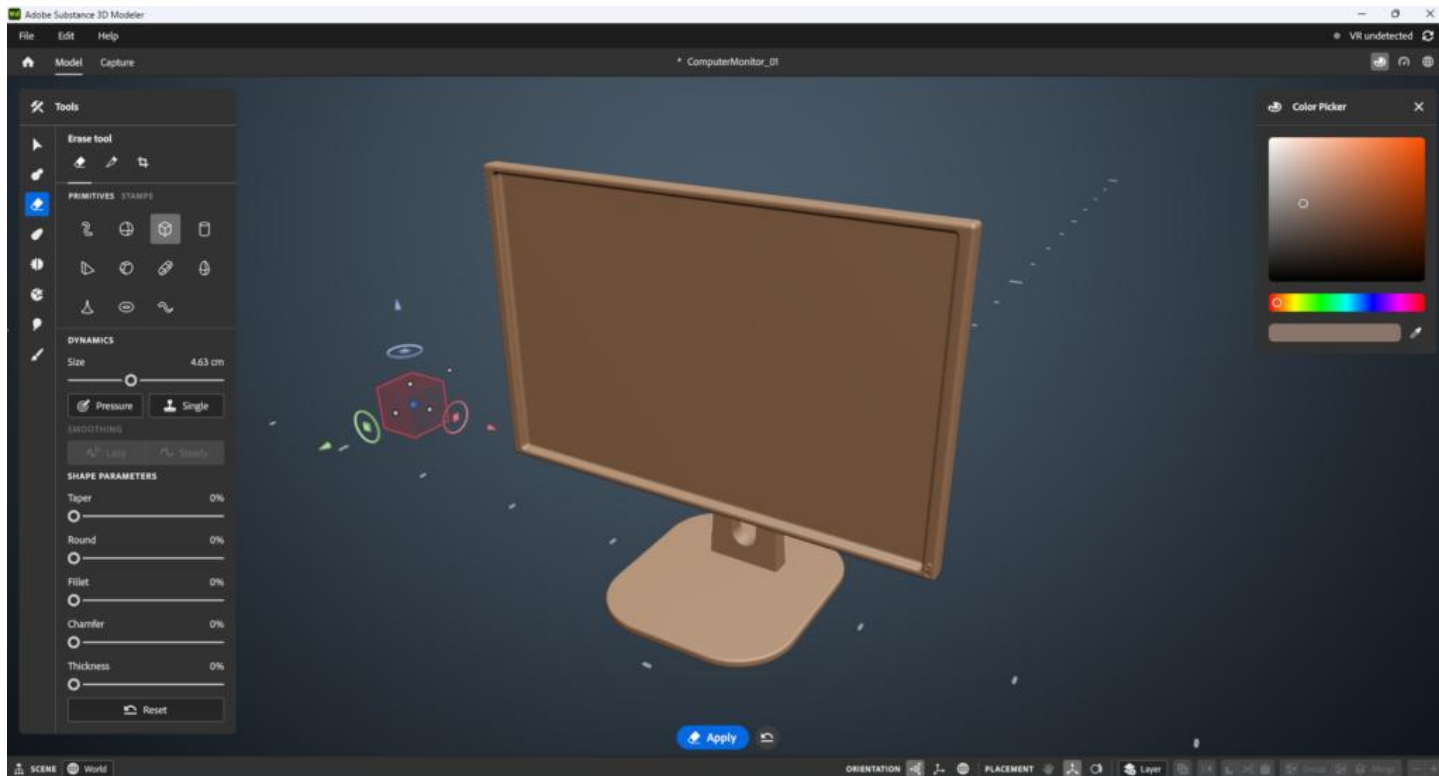
- **Viewport:** Displays the 3D model for viewing, navigation, and editing.
- **Toolbars:** Provide quick access to tools such as brushes, selection, and painting tools.
- **Panels:** Offer settings for properties, layers, materials, assets, and other resources.

Panels can be resized, moved, docked, or floated to create a personalized workspace. Arrange frequently used panels near the viewport to reduce interruptions and improve workflow efficiency.



Gaming

Workspace Layout and Panel Organization



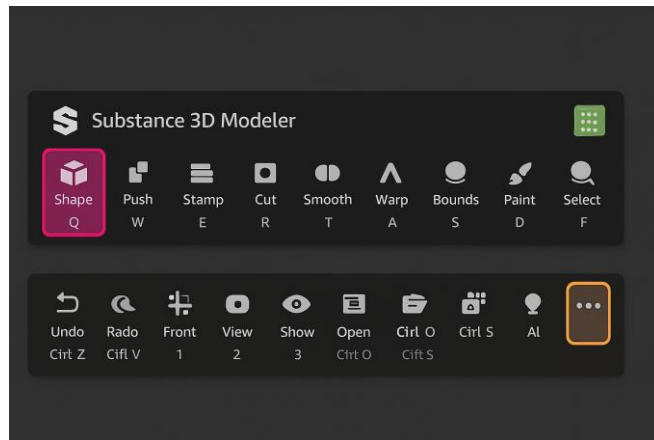


Toolbar and Shortcut Bar Functions

Toolbar and Shortcuts

The toolbar or palette provides quick access to frequently used tools, including sculpting, painting, erasing, and selection tools. Many tools can also be activated directly with keyboard shortcuts, such as C for Clay, E for Erase, and V for Select.

Using shortcuts and arranging the interface through **File > Preferences > Desktop** reduces time spent searching for commands and supports a smoother workflow.





Viewport Controls and Viewing Modes

The viewport is the main area for viewing and working on a 3D model.

- **3D View:** Displays the model with its materials, textures, and lighting. It is used to inspect surface details and paint directly on the mesh.
- **2D View:** Shows the model's UV islands, allowing precise texture painting and editing in UV space.
- **Split View:** Displays both 2D and 3D views together for easier texture alignment.



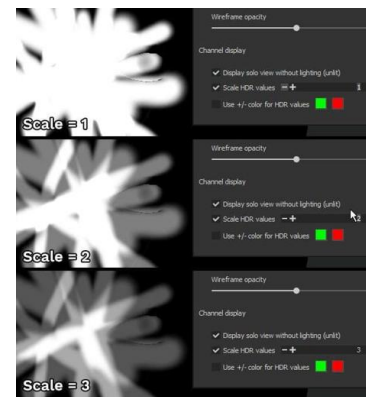


Viewport Controls and Viewing Modes

Viewport controls help you inspect and work on the model more efficiently.

- **Reset View:** Returns the camera to its default position.
- **Fit to Selection:** Centers and zooms the viewport on the selected object or element.
- **View Settings:** Adjust display options such as material shading, wireframe visibility, and lighting for clearer inspection.

Switching viewport modes and display settings helps optimize the workflow for modeling, texture painting, and detail inspection. Substance Painter's Viewport menu also includes options to display materials and center the camera on the entire mesh.





Gaming

TASK



Sculpt a Rough Stone Surface



Gaming

UI and navigation (VR/Desktop modes)

Shambavi BN

Department of Computer Applications



UI and Navigation (VR/Desktop modes)

- Explore the user interface and navigation techniques available in both VR and Desktop modes.
- Understand essential tools and spatial controls that simplify 3D creation and modeling workflows.
- Learn interactive approaches for improved productivity in digital design with Substance 3D.



Substance 3D Desktop User Interface Overview

The desktop UI is designed with modular panels to enhance workflow efficiency:

- Top Bar: Contains menus for file operations, editing, and view options.
- Toolbar: Provides quick access to core tools such as selection, transform, brush, and navigation.
- Viewport: The central 3D workspace where models and scenes are viewed and manipulated.
- Side Panels: Include context-sensitive panels like Properties, Scene hierarchy, Layers, and Material settings for detailed control.

Panels can be docked, collapsed, or resized for personalized workspace setup. The layout supports multitasking by allowing multiple viewports and split-screen views for comparing different project aspects.



Substance 3D VR User Interface Overview

In VR mode, the UI adapts to a 3D spatial environment, allowing natural interaction using VR controllers.

- **Floating Panels:** Contextual menus and tool palettes appear as movable panels within the VR space.
- **Controller-Based Interaction:** Users select, move, scale, and rotate objects using handheld VR controllers with intuitive gestures.
- **Spatial Awareness:** The UI supports immersive exploration with 360-degree room-scale navigation.
- **Tool Access:** Quick access menus are designed around controller buttons and thumbsticks for efficiency.

VR UI minimizes clutter by embedding tools into the environment, enhancing focus on modeling and sculpting tasks.



Core Panels and Tools

- **Scene Panel:** Manages all objects and assets in the 3D scene. Users can select, hide, or organize items hierarchically for easy workflow navigation.
- **Properties Panel:** Displays detailed parameters and attributes of the selected object, such as position, scale, rotation, material properties, and appearance adjustments.
- **Actions Menu:** Contains essential commands and shortcuts for modeling, painting, texturing, and scene composition, aiding in quick execution of common tasks.

These panels are integral to both Desktop and VR modes, adapting their layout according to the workspace but maintaining core functionality for consistency.



Navigation in Adobe Substance 3D Desktop Mode

Camera Controls:

- Orbit: Rotate the camera around the focal point by dragging the mouse with the right button or using specific keys.
- Pan: Move the view horizontally or vertically without changing the camera angle, usually by dragging with the middle mouse button or holding Shift + left button.
- Dolly (Zoom): Zoom in and out using the mouse wheel or keyboard shortcuts to focus on details or broader views.



Navigation in Adobe Substance 3D Desktop Mode

Viewport Interaction:

- Use gizmos (move, rotate, scale) to manipulate objects precisely within the 3D view.
- Selection and transformation tools are accessible from toolbars or shortcut keys.

Support for multiple viewports, perspective and orthographic views, and snapping options for precise modeling.



Navigation in Adobe Substance 3D VR Mode

3D Spatial Navigation:

Move naturally within the VR workspace using room-scale movement or controller joystick for locomotion.

Controller Gestures:

- Use VR controllers to grab, move, scale, and rotate objects intuitively with hand gestures and button presses.
- Pinch, grab, or point gestures facilitate realistic manipulation of 3D assets in the virtual environment.



Navigation in Adobe Substance 3D VR Mode

Axis Rotation and Scaling:

Precise control over objects by activating gizmos attached to objects via controllers, enabling uniform or axis-specific transformations.

Menu Access and Tool Selection:

Floating menus and tool palettes appear within the VR space for easy access without breaking immersion.



Switching Between Desktop and VR Modes

Mode Transition:

- Start Substance 3D in Desktop mode for standard 3D modeling with mouse and keyboard.
- Connect VR hardware (such as Oculus, HTC Vive) and launch VR mode within the software.
- Use in-app menu options or shortcuts to toggle between Desktop and VR modes seamlessly.



Switching Between Desktop and VR Modes

Supported VR Hardware:

- Compatible with major VR headsets supporting room-scale tracking.
- Ensure VR drivers and software are updated for smooth integration.

Workflow Integration:

- Save projects in desktop mode, then open in VR for immersive editing.
- Changes made in one mode reflect immediately in the other for continuous workflow.



Workspace Customization

- Panel Management: Panels like Properties, Scene Hierarchy, Layers, and Materials can be docked, undocked, collapsed, or resized according to user preference.
- Layout Personalization: Users can create custom workspace layouts tailored for specific tasks, such as texturing, sculpting, or rendering.
- Toggle Visibility: Quickly show or hide UI elements to reduce clutter and focus on the current workflow step.
- Multi-Viewport Support: Open and arrange multiple viewports to see different angles, orthographic vs perspective views, or various parts of the project simultaneously.



Advanced Navigation

- Fit to Selection:

Quickly frame the selected object or group in the viewport for focused editing by pressing a shortcut (e.g., F key).

- Reset View:

Return the camera to the default position and orientation to regain a standard workspace view.

- Perspective/Orthographic Toggle:

Switch between perspective view (realistic 3D depth) and orthographic view (parallel projection) for technical accuracy.

- Ray Tracing Mode:



- Ray Tracing Mode:

Enable real-time ray tracing for accurate lighting and reflections during rendering previews, enhancing visual feedback.

- Gizmo Snapping:

Use snapping options on movement, rotation, and scaling gizmos for precise adjustments aligned to grids or specific increments.



Gaming

**Modeling with clay-like tools (Add,
Subtract, Smooth)**

Shambavi BN

Department of Computer Applications



Gaming

Clay-like Tools in Modeling

Clay-like tools simulate traditional sculpting by adding or removing material, providing intuitive control over model shapes.

They allow artists to shape digital models naturally, enhancing creativity and precision.



Overview of Clay-like Tools

What Are Clay-like Tools?

Digital sculpting brushes that simulate working with physical clay.
Essential for shaping and refining 3D models with organic, fluid strokes.



Add



Subtract



Smooth



Overview of Clay-like Tools

Add Tool: Adds virtual clay or material to the mesh, building volume.

Subtract Tool: Removes or carves material from the mesh for hollowing or details.

Smooth Tool: Evenly blends and smooths transitions on the model surface for refined shapes.

Their Role in Workflow:

Establish base forms with Add and Subtract.

Refine and polish shapes using Smooth for a natural look.



Functionality and Use Cases:

- The Add tool works like applying clay onto your model, building up volume and form.
- Ideal for creating bulges, muscle masses, or expanding surfaces in organic models.

How to Use in Substance 3D:

- Select the Add brush from the sculpting tools panel.
- Adjust brush size and intensity to control the amount of material added.
- Click and drag on the mesh to “paint” on volume.



The Subtract Tool

Functionality and Use Cases:

- The Subtract tool is the opposite of Add; it removes virtual clay from the mesh.
- Used to carve details, create indentations, or hollow out parts of a model.

How to Use in Substance 3D:

- Select the Subtract brush in the sculpting toolset.
- Adjust brush size and strength to control how much material is removed.
- Click and drag to carve into the surface, revealing deeper forms.



Functionality and Use Cases:

- The Smooth tool evens out surfaces by blending together rough or uneven areas.
- Essential for polishing details and creating natural transitions between added and subtracted forms.

How to Use in Substance 3D:

- Select the Smooth brush from the sculpting tools.
- Adjust brush size and intensity to control the degree of smoothing.
- Click and drag over the model surface to soften harsh edges and blend details.



Gaming

Symmetry, layers, and voxel concepts

Shambavi BN

Department of Computer Applications



Understanding Symmetry in Substance 3D

What is Symmetry?

Symmetry in 3D texturing means mirroring or repeating brush strokes or effects across one or more axes to create balanced and consistent designs.

Importance of Symmetry:

- Speeds up the texturing process by duplicating work automatically
- Ensures uniformity and precision in design, especially for characters and objects
- Helps create complex patterns easily



Symmetry Tools & Types in Substance 3D

Types of Symmetry:

- Mirror Symmetry: Reflects painting or sculpting across a central axis (X, Y, or Z).
- Radial Symmetry: Repeats strokes around a center point in a circle, useful for patterns and mandalas.



How to Enable Symmetry:

- Access symmetry settings in the brush or sculpt tool panel.
- Choose the axis or radial count based on the design needs.



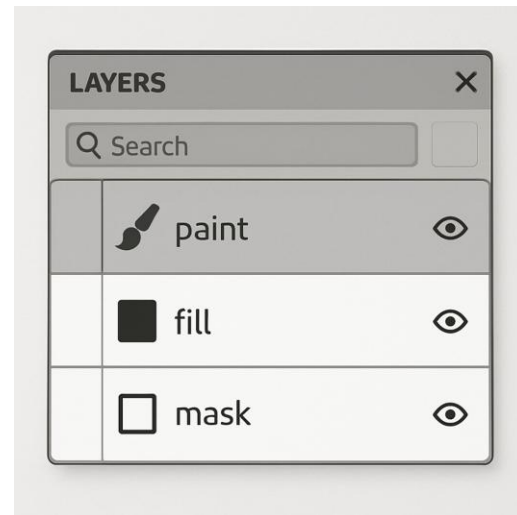
Layer System in Adobe Substance 3D

What Are Layers?

Layers work like transparent sheets stacked on top of each other, each holding different textures, effects, or adjustments applied to the model.

Importance of Layers:

- Enable non-destructive editing
- Separate different texturing elements for better control
- Facilitate easy tweaking without affecting the entire model





Using and Managing Layers in Substance 3D

Layer Blending Modes:

- Control how layers interact visually (e.g., Multiply, Overlay, Add)
- Useful for adding depth, shadows, highlights, and surface details

Layer Organization:

- Group related layers for better workflow management
- Rename layers logically for easy identification

Non-destructive Workflow:

- Use masks and adjustment layers to refine textures without permanent changes
- Adjust opacity and blending anytime for fine-tuning results



Introduction to Voxels in Substance 3D

What Are Voxels?

Voxels are volumetric pixels, representing values in 3D space instead of flat 2D pixels. Think of them as tiny cubes or blocks that make up a 3D object.

Difference from Traditional 3D Geometry:

Traditional 3D models use polygons (faces and vertices) to define shapes. Voxels represent 3D space as a grid of volumetric data, allowing more flexible sculpting and texturing.



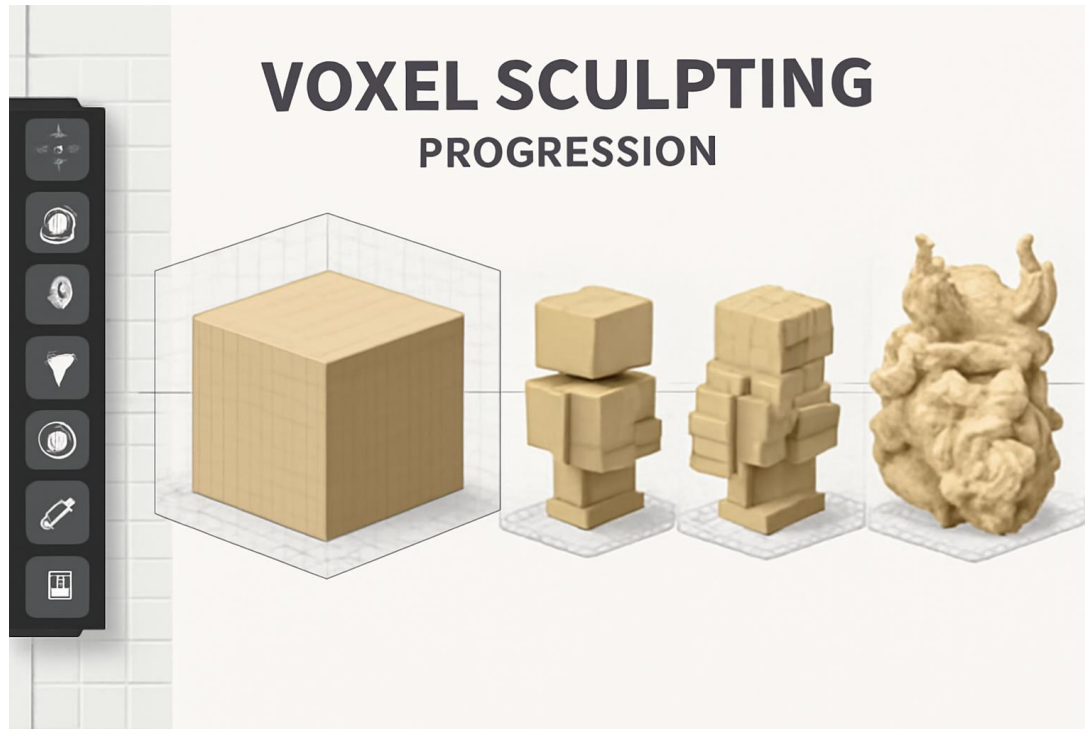
Introduction to Voxels in Substance 3D

Why Use Voxels?

Capture fine details with more natural, organic forms.

Easier to add or subtract material without worrying about mesh topology.

Great for sculpting complex surfaces and shapes.





Voxel Workflow in Substance 3D

Voxel-Based Sculpting:

- Start with a voxel object to sculpt and add details freely without worrying about mesh stretching or topology mistakes.
- Allows dynamic changes like adding or subtracting volume in a natural way.

Advantages of Voxels:

- Smooth and seamless detailing for complex organic shapes.
- Easy blending of shapes and textures.
- Supports complex layering of materials on a volumetric basis.

Voxel to Polygon Conversion:

- After sculpting in voxels, models can be converted to polygon meshes for further use in animation or game engines.



Voxel Workflow in Substance 3D

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Gaming

**Hard surface vs. organic modelling,
Exporting models (to Painter or
Stager)**

Shambavi BN

Department of Computer Applications



Definition of Hard Surface Modelling

Hard Surface Modelling is focused on creating objects that have sharp edges, clean lines, and flat surfaces.

Commonly used for mechanical and man-made objects such as vehicles, weapons, gadgets, robots, and architectural designs.

It involves precise geometric shapes like cubes, cylinders, and planes often combined using techniques like Boolean operations.

Emphasis is on hard edges, bevels, and crisp surface details.





Definition of Organic Modelling

Organic Modelling focuses on creating natural, flowing, and curved shapes that mimic living organisms.

Common subjects include humans, animals, plants, and creatures with irregular, soft, and asymmetrical forms.

It relies heavily on sculpting tools that allow pushing, pulling, and smoothing surfaces to achieve lifelike details.

Emphasizes smooth transitions, volume, and complex surface anatomy.





Differences Between Hard Surface and Organic Modelling

Feature	Hard Surface Modelling	Organic Modelling
Shape	Geometric, sharp, angular	Rounded, smooth, flowing
Subject Matter	Mechanical, man-made objects	Living organisms and natural forms
Topology	Clean, often uses quads and n-gons	Dense, flowing edge loops for natural detail
Tools & Techniques	Boolean operations, hard edges, bevels	Sculpting brushes, push/pull, dynamic topology
Surface Texture	Hard, metallic, plastic-like surfaces	Soft, skin, fur, organic textures



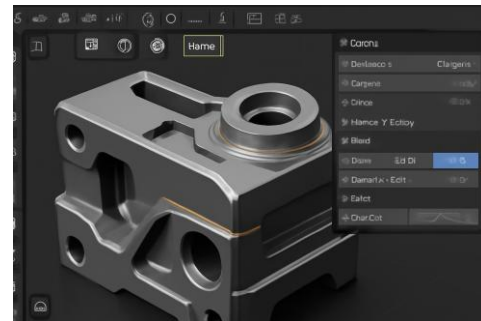
Techniques and Tools for Hard Surface Modelling

Boolean Operations: Used to combine, subtract, or intersect shapes to create complex hard surface details.

Edge Creasing and Beveling: Techniques to maintain sharp edges and add realism. Beveling helps catch light and shows wear.

Subdivision Surface Modelling: Smooths the mesh while preserving sharp edges using control loops.

CAD and Poly Modelling Tools: Software like Autodesk Maya, 3ds Max, Blender, and CAD tools are commonly used.





Techniques and Tools for Organic Modelling

Sculpting Tools: Use brush-based tools to push, pull, smooth, and shape the mesh dynamically, allowing intuitive creation of natural forms.

Dynamesh and Remeshing: Techniques that dynamically adjust mesh topology to maintain consistent detail while sculpting.

Subdivision for Detail: Adding multiple levels of subdivision to refine surface details and create highly detailed models.

Topology Retopology: After sculpting, creating cleaner, animation-friendly topology for use in games or films.

Common software includes ZBrush, Blender Sculpt Mode, Mudbox.



Topology Considerations in Hard Surface and Organic Modelling

Topology refers to the arrangement of polygons in a 3D mesh. Good topology ensures clean, efficient, and deformation-friendly models.

Hard Surface Modelling: Emphasizes clean, sharp edges, often uses quads with supporting edge loops to maintain shapes.

Organic Modelling: Focus on edge flow and loops that follow muscle and anatomical forms for smooth deformation.

Importance: Proper topology improves animation, texturing, and rendering quality.

Tools exist to check and optimize topology before export or rigging. Compiled by Shambavi BN



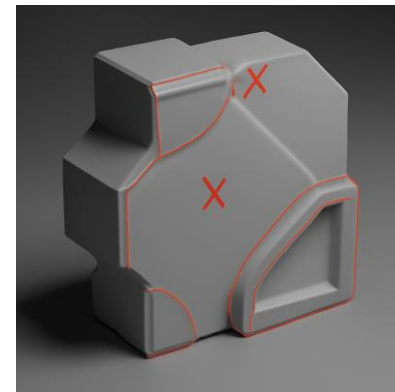
Challenges in Hard Surface Modelling

Maintaining sharp, clean edges without distorting the mesh during subdivision and smoothing.

Avoiding geometry errors such as non-manifold edges and ngons that can cause issues in rendering or exporting.

Balancing mesh density to keep the model lightweight while preserving details.

Handling complex shapes where Boolean operations can create messy topology requiring cleanup.





Challenges in Organic Modelling

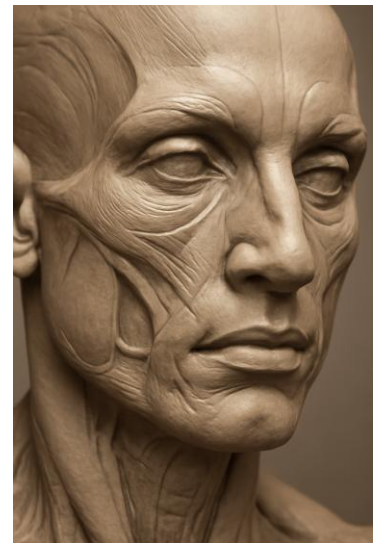
Achieving natural, lifelike curves and forms that match anatomy.

Creating smooth transitions between body parts without visible seams or distortions.

Managing high polygon counts needed for detailed sculpting while keeping performance manageable.

Retopologizing complex sculpts into clean, animation-friendly meshes.

Capturing fine surface details like skin pores, wrinkles, and muscle striations without overloading the mesh.





Gaming

Exporting models (to Painter or Stager)

Shambavi BN

Department of Computer Applications



Complete Workflow Overview

Typical 3D asset workflow:

1. Create or obtain a 3D model
2. Clean and optimize the geometry
3. Create UV maps
4. Export the model in a compatible format
5. Import the model into Substance 3D Painter
6. Bake mesh maps and paint materials
7. Export texture maps from Painter
8. Import the model and textures into Substance 3D Stager
9. Add lights, environment, camera, and render output



Painter vs Stager

Feature	Substance 3D Painter	Substance 3D Stager
Main purpose	Painting textures and materials on a model	Building, lighting, and rendering 3D scenes
Main input	UV-mapped 3D mesh	Models, materials, lights, images, and environments
Typical output	Texture maps such as Base Color, Roughness, Normal, Metallic	Rendered product image, scene render, or staged composition
Important requirement	Clean geometry and good UV layout	Correct scale, material assignments, and scene-ready model
Example use	Paint scratches on a game sword	Place the sword on a table with studio lighting



Preparing the Model Before Export

Before export, check the following:

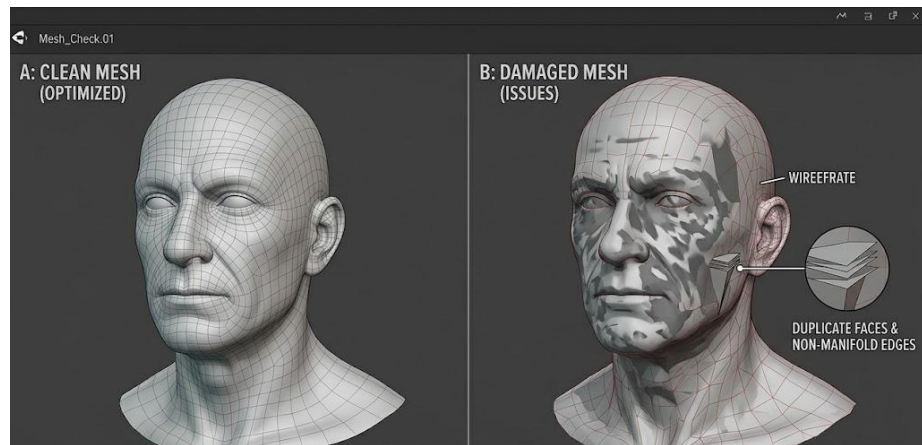
- Remove hidden, unused, or duplicate objects.
- Delete unwanted faces inside the model.
- Remove duplicate vertices, edges, and faces.
- Check for non-manifold geometry.
- Verify that surface normals face outward.
- Apply transforms such as scale, rotation, and position if required by the modeling application.
- Ensure separate parts have meaningful names.
- Assign material IDs or material slots to separate areas where different textures are needed.



Preparing the Model Before Export

Why this matters:

- Bad geometry can create baking errors, texture seams, black areas, visual artifacts, and performance issues.
- Painter documentation specifically recommends avoiding very small triangles, invalid normals, non-manifold geometry, and duplicated mesh elements.





UV Mapping for Substance Painter

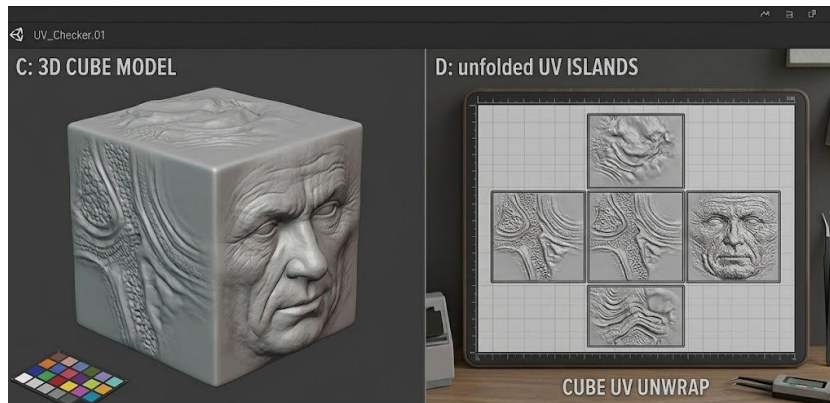
- A UV map tells Painter where each painted pixel belongs on the 3D model.
- UV islands are flattened sections of the model.
- The usual texture space ranges from 0 to 1 in both U and V directions.
- UV seams are locations where the 3D surface is cut and unfolded.
- Texel density means the amount of texture detail per unit of model surface.



UV Mapping for Substance Painter

Good UV practices:

- Avoid overlapping UV islands unless the overlap is intentional.
- Keep UV islands inside the 0–1 UV space for standard workflows.
- Avoid stretched UV islands.
- Maintain consistent texel density.
- Place seams in less visible regions where possible.





Choosing an Export Format

Common 3D model formats include:

Format	Best use	Key point
OBJ	Simple geometry transfer	Widely supported; may require separate material and texture files
FBX	Models with hierarchy, materials, and animation data	Good for many production pipelines
GLTF/GLB	Web and real-time 3D workflows	Modern format; GLB stores content in one binary file
USD/USDZ	Advanced scene exchange and AR-oriented workflows	Useful in modern Adobe and AR pipelines
PLY	Mesh and scan-based assets	Often used for geometry-focused assets



Gaming

Choosing an Export Format





Export Settings Checklist

Use this checklist before clicking Export:

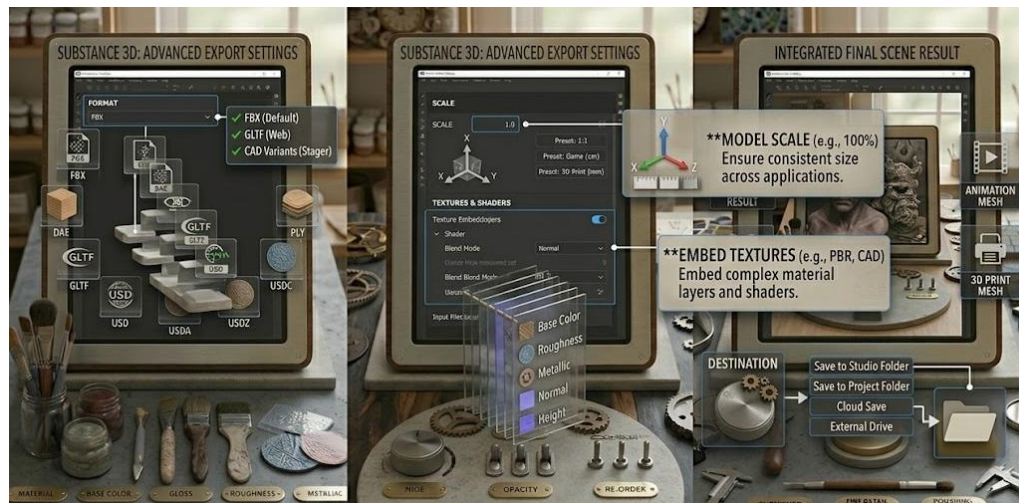
- Export only the required model objects.
- Confirm the correct file format.
- Use meaningful file names.
- Set correct measurement units.
- Apply or freeze transforms where appropriate.
- Check model orientation: up-axis and forward-axis.
- Include materials only when needed.
- Include texture files if the format and workflow support them.
- Keep texture files in an organized folder.
- Avoid spaces, special characters, and unclear names in file paths.
- Save a backup copy of the original modeling file.



Export Settings Checklist

Example naming convention:

- GameSword_Low.fbx
- GameSword_High.fbx
- GameSword_Textures
- GameSword_Painter.spp
- GameSword_Stager.ssg





Exporting a Model to Substance Painter

Steps to prepare a model for Painter:

1. Complete the low-poly or final mesh.
2. Check geometry, normals, and UVs.
3. Export as OBJ, FBX, GLTF/GLB, or USD-based format.
4. Open Substance 3D Painter.
5. Choose File > New.
6. Select the exported mesh file.
7. Set document resolution, such as 2048×2048 or 4096×4096 .
8. Import texture sets if they already exist.
9. Create the project.

Painter identifies mesh parts through texture sets, often based on material assignments in the original model.



Exporting a Model to Substance Painter

Important:

- Use a low-poly mesh for painting.
- Keep a high-poly model available if you want to bake details such as normal maps, ambient occlusion, curvature, and thickness maps.



Baking Maps in Painter

Baking transfers surface information from a high-resolution model to a low-resolution model.

Common baked maps:

- Normal map: Simulates small high-detail surface forms.
- Ambient occlusion map: Adds contact shadows in creases and corners.
- Curvature map: Detects edges and cavities.
- World space normal map: Stores face direction in world space.
- Position map: Stores position information across the model.
- Thickness map: Helps generate effects such as subsurface or edge wear.



Baking Maps in Painter

Why baking is important:

- Smart materials and generators depend on baked maps.
- A clean bake improves realism and reduces manual work.
- Naming high-poly and low-poly parts correctly helps prevent projection errors.

Example:

A low-poly helmet can receive bolts, dents, edge wear, and grooves from a high-poly helmet through baked maps.



Baking Maps in Painter





Importing Models into Stager

To import a model into Stager:

1. Open a new or existing Stager project.
2. Choose File > Import > 3D Model.
3. Browse to the exported model file.
4. Select the model and open it.
5. Review import settings.
6. Place the model in the viewport.
7. Check scale, orientation, and material assignment.
8. Save the Stager project as an .ssg file.

Stager is intended for staging 3D objects in a scene, including models, materials, lights, cameras, and background images. Its project file extension is .ssg



Applying Painter Textures in Stager

After importing the model into Stager:

- Select the model in the Scene panel.
- Select the required material slot.
- Open material properties.
- Connect exported Painter texture maps to the correct channels.



Applying Painter Textures in Stager

Typical map connections:

Painter Map	Stager Material Channel
Base Color	Base Color
Roughness	Roughness
Metallic	Metallic
Normal	Normal
Opacity	Opacity
Emissive	Emissive
Height	Height or Displacement, when supported



Applying Painter Textures in Stager

Important rule:

- Do not connect a roughness map to the metallic channel.
- Do not use a normal map as a color image.
- Verify whether the rendering workflow uses roughness or glossiness before connecting maps.



Applying Painter Textures in Stager

Problem	Likely cause	Solution
Model appears black in Painter	Flipped or invalid normals	Recalculate normals outward in the modeling software
Texture paint appears distorted	Poor or stretched UVs	Improve UV unwrap and reduce stretching
Texture appears in the wrong place	Overlapping UVs or incorrect UV layout	Separate islands or correct UV placement
Model is too large or too small in Stager	Unit mismatch	Set units before export; rescale during import if needed
Materials are missing	Texture paths are broken or maps were not imported	Keep texture files organized and reconnect maps



Applying Painter Textures in Stager

Problem	Likely cause	Solution
Bake has artifacts	Incorrect high-poly/low-poly alignment or bad geometry	Use cages, correct naming, and clean mesh topology
Seams are visible	UV seams, inconsistent texel density, or paint mismatch	Place seams carefully and paint across seam regions
Stager render looks unrealistic	Missing environment, weak lighting, or incorrect roughness	Add HDRI or lights and verify material maps



THANK YOU

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